

Capture the Flag 6: Active or Inactive?

Goal: Make your Student Directory respond to student data by conditionally showing, hiding, and changing UI elements based on each student's state.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding conditional UI.

Create:

`feature/conditional-ui`

Screenshot:

```
MINGW64; c:/Users/User/Documents/Moreno_A6/IT314AB_Moreno-A
$ git branch
* feature/student-interactions
  main

User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/student-interactions)
$ git status
On branch feature/student-interactions
Your branch is up to date with 'origin/feature/student-interactions'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  IT314AB_Moreno-A/

nothing added to commit but untracked files present (use "git add" to track)

User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/student-interactions)
$ git checkout -b feature/conditional-ui
Switched to a new branch 'feature/conditional-ui'

User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/conditional-ui)
$ git branch
* feature/conditional-ui
  feature/student-interactions
  main

User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/conditional-ui)
$ |
```

🚩 Flag 1 — Show Active or Inactive

Objective: Change the Student Card based on whether a student is active or inactive.

Add an **active** status to your student data.

Research how Flutter uses **conditional expressions** to display different widgets.

Example states:

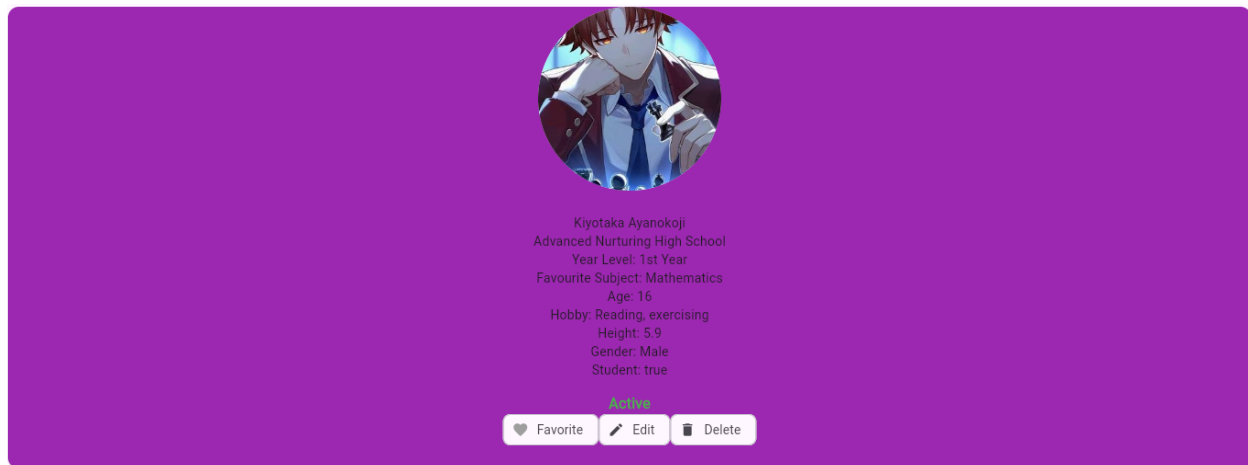
- Active
- Inactive

Test

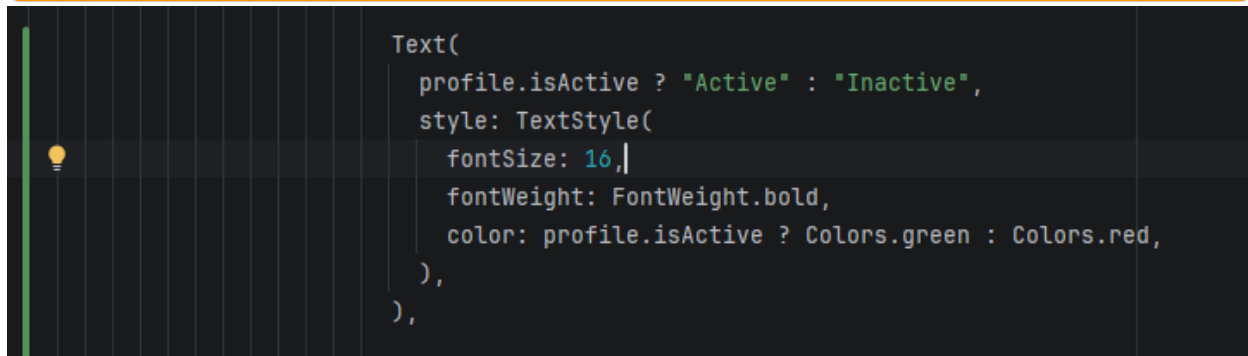
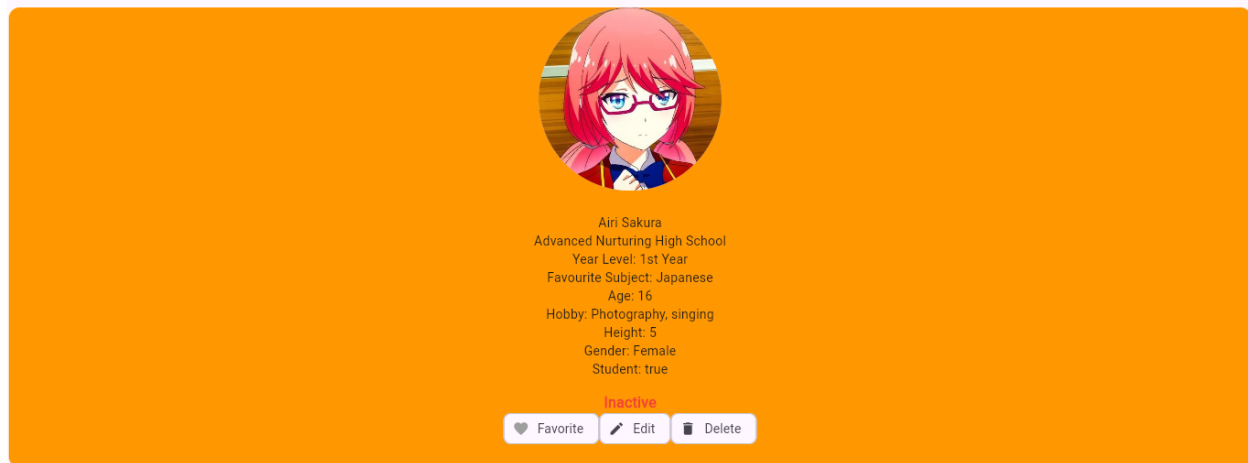
Active students should display an Active indicator.

Inactive students should display an Inactive indicator.

Screenshot:



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🚩 Flag 2 — Show or Hide

Objective: Display certain UI elements only when a condition is true.

Add a conditional element to the Student Card.

Examples:

- Favorite badge appears only when favorited

- Warning appears only when inactive
- Delete action appears only if active

Research ways to conditionally include or exclude widgets from the widget tree.

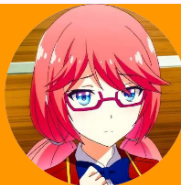
Test

When the condition is true, the element appears.

When the condition is false, the element disappears.

Screenshot:

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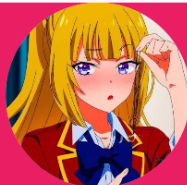


Airi Sakura
Advanced Nurturing High School
Year Level: 1st Year
Favourite Subject: Japanese
Age: 16
Hobby: Photography, singing
Height: 5
Gender: Female
Student: true

Inactive

Warning: This student is inactive

Edit Delete

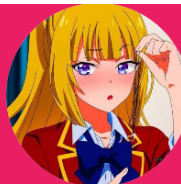


Kei Karuizawa
Advanced Nurturing High School
Year Level: 1st Year
Favourite Subject: English
Age: 16
Hobby: Shopping, socializing
Height: 5
Gender: Female
Student: true

Active

Favorite Student

Favorite Edit



Kei Karuizawa
Advanced Nurturing High School
Year Level: 1st Year
Favourite Subject: English
Age: 16
Hobby: Shopping, socializing
Height: 5
Gender: Female
Student: true

Active

Favorite Edit

```

if (profile.isFavorite)
  const Text(
    "Favorite Student",
    style: TextStyle(
      fontSize: 16,
      fontWeight: FontWeight.bold,
      color: Colors.yellow,
    ),
  ),
),

```

```

if (!profile.isActive)
  const Text(
    "Warning: This student is inactive",
    style: TextStyle(
      fontWeight: FontWeight.bold,
      color: Colors.red
    ),
  ),
),

```

```

const SizedBox(height: 10),
if (!profile.isActive)
InputChip(
  avatar: const Icon(
    Icons.delete,
    size: 18,
  ),
  label: const Text("Delete"),
  onPressed: () {
    setState(() {
      profiles.removeAt(index);
    });
  },
),

```

🚩 Flag 3 — The Empty Directory

Objective: Handle situations where there are no students to display.

Modify your Student Directory so it detects when the student list is empty.

Create an **empty state**.

Examples:

```
167         body: profiles.isEmpty
168           ? const Center(
169             child: Text(
170               "No student found",
171               style: TextStyle(fontSize: 25),|
172             ),
173           )
174         : ListView.builder(
```

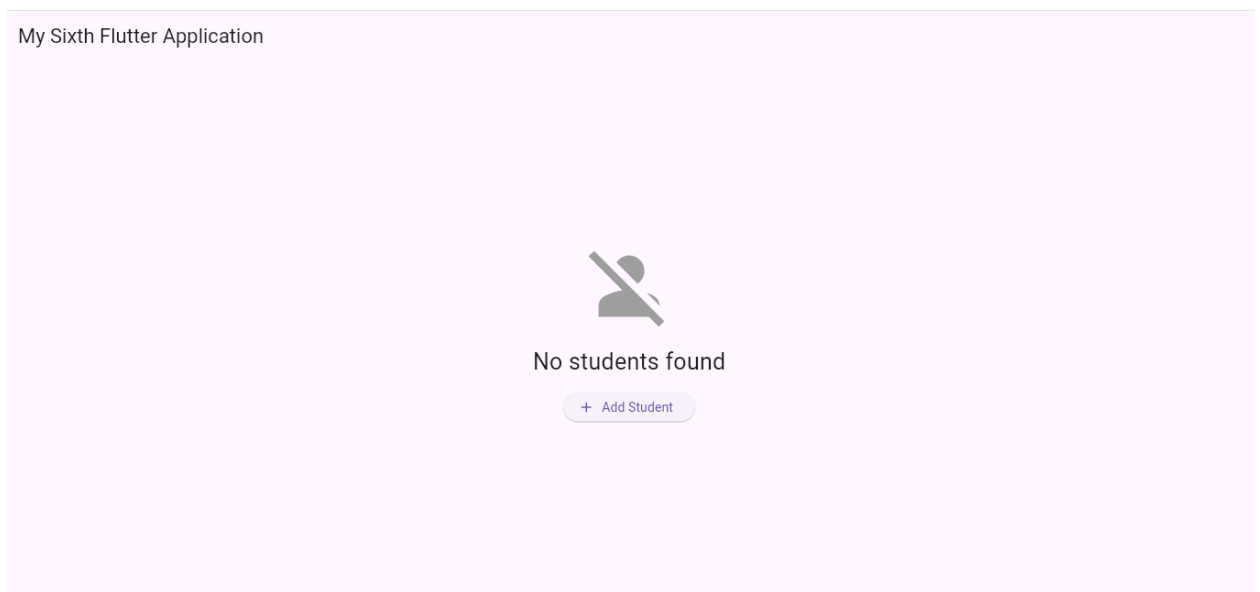
- Empty-state icon
- Button to add a student

Test

Delete all students in your application (not the code)

Instead of showing a blank screen, your application should display your empty state.

Screenshot:



```

body: profiles.isEmpty
  ? Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        const Icon(
          Icons.person_off,
          size: 100,
          color: Colors.grey,
        ),
        const SizedBox(height: 15),
        const Text(
          "No students found",
          style: TextStyle(fontSize: 25),
        ),
      ],
    ),
  ),

```

```

Deleted Kakeru Kyuen
Deleted Kei Karuizawa
Deleted Kikyo Kushida
Deleted Kiyotaka Ayanokoji
Deleted Suzune Horikita
Add Student pressed
Add Student pressed

```

🚩 Flag 4 — Loading the Directory

Objective: Display different UI while student data is being loaded.

Create a loading state for your Student Directory.

Research how Flutter can conditionally display a loading indicator.

Examples:

- `CircularProgressIndicator`
- Loading message
- Loading placeholder

Test

When the application is in a loading state, the student list should not be displayed as the main content.

Once loading is complete, the Student Directory should appear.

Screenshot:

My Sixth Flutter Application

Loading students...

My Sixth Flutter Application



Airi Sakura
Advanced Nurturing High School
Year Level: 1st Year
Favourite Subject: Japanese
Age: 16
Hobby: Photography, singing
Height: 5
Gender: Female
Student: true

Inactive
Warning: This student is inactive

Edit Delete

```
class _MyAppState extends State<MyApp> {  
  bool isLoading = true;  
  
  @override  
  void initState() {  
    super.initState();  
    Future.delayed(const Duration(seconds: 5), () {  
      setState(() {  
        isLoading = false;  
      });  
    });  
  }  
}
```

```

body: isLoading
  ? const Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        CircularProgressIndicator(),
        SizedBox(height: 15),
        Text(
          "Loading students",
          style: TextStyle(fontSize: 20),
        ),
      ],
    ),
  )
: profiles.isEmpty
  ? Center(

```

🚩 Flag 5 — Multiple States

Objective: Make your Student Directory handle multiple UI states correctly.

Your application should now be able to display:

1. Loading
2. Students available
3. No students available

Determine which state should be displayed based on your data.

Test

Verify each state:

Loading

→ Loading UI

Students available

→ Student List

Empty list

→ Empty State

Only the appropriate UI should appear at a time.

Screenshot:

My Sixth Flutter Application



Loading students...

My Sixth Flutter Application



Kakeru Ryuen
Advanced Nurturing High School
Year Level: 1st Year
Favourite Subject: Social Studies
Age: 17
Hobby: Fighting, strategizing
Height: 5'9
Gender: Male
Student: true

Inactive

Warning: This student is inactive

Edit

Delete



My Sixth Flutter Application



No students found

+ Add Student

🚩 Flag 6 — States Work Together

Objective: Make sure conditional UI does not break your existing interactions.

Test the combination of:

- Favorite
- Delete
- Edit
- Active/Inactive
- Show/Hide
- Empty state
- Loading state

Test Scenario

1. Favorite a student.
2. Change or display their Active/Inactive state.
3. Delete students until the directory becomes empty.
4. Verify the Empty State appears.
5. Reload or restore students.
6. Verify the Student Directory appears again.

The application should continue functioning correctly as its state changes.

Screenshot:

```
Favorite pressed for Kei Karuizawa
Deleted Kei Karuizawa
Deleted Airi Sakura
Deleted Kakeru Ryuen
Student Card tapped: Kikyo Kushida
Student Card tapped: Kikyo Kushida
Deleted Kikyo Kushida
Deleted Kiyotaka Ayanokoji
Student Card tapped: Suzune Horikita
Student Card tapped: Suzune Horikita
Deleted Suzune Horikita
Add Student pressed
```



Kei Karuizawa
Advanced Nurturing High School
Year Level: 1st Year
Favourite Subject: English
Age: 16
Hobby: Shopping, socializing
Height: 5
Gender: Female
Student: true

Active

Favorite Student

Favorite Edit Delete

Edit Student

Edit Kei Karuizawa?

Cancel

Edit

Hobby: Shopping, socializing

Height: 5

Gender: Female

Student: true

Active

Favorite Student



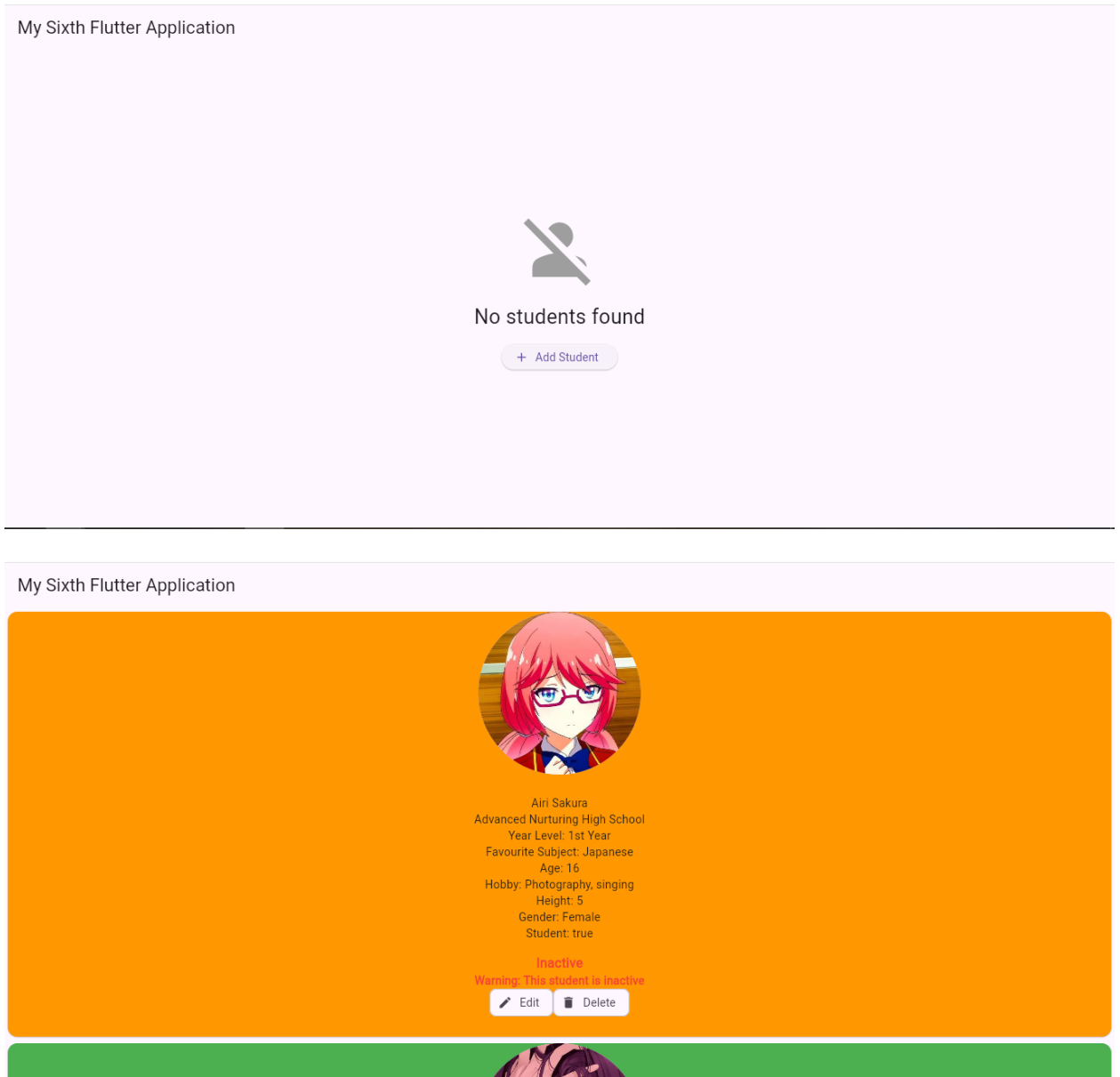
Favorite



Edit



Delete



🚩 Flag 7 — Save Your Progress

Objective: Commit and push your completed Conditional UI Student Directory.

Use a proper Git commit message.

Branch:

`feature/conditional-ui`

Screenshot:

```
User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/conditional-ui)
$ git commit -m "feat: add conditional UI to student directory"
[feature/conditional-ui 34ccd92] feat: add conditional UI to student directory
1 file changed, 90 insertions(+), 8 deletions(-)
```

```
User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/conditional-ui)
$ git push -u origin feature/conditional-ui
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 6 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1.18 KiB | 18.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'feature/conditional-ui' on GitHub by visiting:
remote:   https://github.com/AnthonySatoru/IT314AB_Moreno-A/pull/new/feature/conditional-ui
remote:
To https://github.com/AnthonySatoru/IT314AB_Moreno-A.git
 * [new branch]      feature/conditional-ui -> feature/conditional-ui
branch 'feature/conditional-ui' set up to track 'origin/feature/conditional-ui'.
```

```
User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/conditional-ui)
$
```

```
User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/conditional-ui)
$ git commit -m "feat: add conditional UI to student directory"
[feature/conditional-ui 34ccd92] feat: add conditional UI to student directory
1 file changed, 90 insertions(+), 8 deletions(-)
```

```
User@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A6/IT314AB_Moreno-A (feature/conditional-ui)
$
```